Predicting E-Cigarette Use Among Thai Youth Using Machine Learning: The Role of Mental Health and Risk Behaviors

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Abstract

Introduction: Currently, the usage of e-cigarettes among Thai youth is on a steady and alarming growing trend. According to data from Thailand's National Statistical Office (2022), the number of e-cigarette users has increased by over 10 percent in just one year. The situation demonstrates that e-cigarette use among Thai youths and teens is concerning and rapidly increasing, necessitating close monitoring and legislative action to minimize long-term health implications.

Aim and Methods: To evaluate the performance across five machine learning models in predicting e-cigarette use behavior among Thai youth, including Logistic Regression, Support Vector Machines, Random Forest, Gradient Boosting Machines, and XGBoost, and to examine the key factors influencing e-cigarette use behavior among Thai youth.

Result: The evaluation of the machine learning model's performance indicated that the Random Forest model had the best performance for the ADASYN-balanced dataset, with a sensitivity of 78.95% and an AUC of 0.73. The most significant key factors for e-cigarette smoking behavior, based on Feature Importance Scores, include methods for solving problems when stressed for people who have previously consumed alcohol (12.62%).

Conclusions: Models in the tree-based model group are more suitable compared to those in the linear model group. Random Forest is the strongest model in the tree-based model group, which was balanced using the ADASYN (Adaptive Synthetic Sampling) method and was able to effectively identify the important aspects influencing e-cigarette smoking behavior.

Implications: The study indicates that machine learning-based prediction, specifically Random Forest, can be used to detect risky behaviors that lead to e-cigarette usage, therefore decreasing e-cigarette use among Thai youths.

Keywords: machine learning, e-cigarette smoking behavior, key factors

Introduction

In recent years, the use of electronic cigarettes among Thai youth has increased at an alarming rate. According to a national survey conducted by the National Statistical Office of Thailand in 2022, the number of electronic cigarette users aged 15 years and older increased more than tenfold, rising from approximately 78,742 individuals in 2021 to 709,677 individuals in 2022. A similar trend was observed among adolescents and young adults aged 15–24 years, for whom the number of users increased from 24,050 to 269,533 during the same period. This rapid escalation has led policymakers to characterize the spread of electronic cigarette use among children and youth as a public health crisis (Thai Senate, 2025).

Advances in data science and machine learning have created new opportunities for developing predictive models capable of identifying individuals at risk of electronic cigarette use with high accuracy. For example, a recent study by Wei Fang et al. (2024) employed feature selection techniques and compared multiple machine learning models, including RF, XGBoost, GBM, SVM, and LR to predict current electronic cigarette use among adults in the United States. Their findings demonstrated that the Random Forest model achieved the best performance (Accuracy $\approx$ 0.992, AUC $\approx$ 0.999), with key predictive factors including age, heavy alcohol consumption, and beliefs regarding the harmfulness of electronic cigarettes.

Despite the growing body of international research, evidence on the application of machine learning models to predict electronic cigarette use among adolescents and youth in Thailand remains limited. Differences in sociocultural context, regulatory environments, and behavioral determinants suggest that predictive models developed in other countries may not be directly transferable to the Thai population.

Therefore, this study aims to evaluate and compare the predictive performance of several machine learning models—namely Logistic Regression, Random Forest, Gradient Boosting Machine, XGBoost, and Support Vector Machines—in predicting electronic cigarette use among Thai youth. The primary objective is to identify the model with the highest predictive accuracy and robustness. The findings are expected to provide valuable insights for designing targeted prevention strategies and informing evidence-based public health policies aimed at reducing electronic cigarette use among Thai adolescents and young adults. Furthermore, the resulting predictive framework may serve as a practical tool for future risk assessment and policy development.

Materials and Methods

This study used secondary data from a research project entitled “Impact of Mental Health on Alcohol Consumption Behavior among Adolescents in Bangkok” conducted by Supaknapha et al. (2024). The data were collected through an online questionnaire covering demographic characteristics, alcohol consumption, e-cigarette use, mental health status, and stress-related factors. The study population consisted of Thai youth aged 15–24 years residing in Bangkok. After data cleaning and validation, 698 participants were included in the analysis.

Descriptive statistics were used to summarize the general characteristics of the participants. Preliminary analyses were conducted to examine the associations between participant characteristics and e-cigarette use. For predictive modeling, the dataset was randomly split into training set and testing set at a ratio of 80:20. Several machine learning models were used to predict e-cigarette, including Logistic Regression, Support Vector Machines, Random Forest, Gradient Boosting Machines, and Extreme Gradient Boosting. Before model training, data preprocessing steps such as feature scaling and feature selection using the Least Absolute Shrinkage and Selection Operator (LASSO) were performed to improve model performance and reduce multicollinearity.

Due to an imbalance in the outcome variable, resampling techniques were applied to address this issue. Adaptive Synthetic Sampling (ADASYN), Synthetic Minority Over-sampling Technique (SMOTE), and Random Over-sampling (ROS) were used to increase the number of samples in the minority class. Model training and resampling procedures were performed using 5-fold stratified cross-validation to enhance model reliability and minimize overfitting.

Model performance was evaluated using multiple metrics, including accuracy, precision, sensitivity, specificity, F1-score, and the area under the receiver operating characteristic curve (AUC). The performance of all models was compared to identify the most suitable predictive approach. In addition, feature importance analysis was conducted for the best-performing model to determine key factors associated with e-cigarette use among Thai youth.

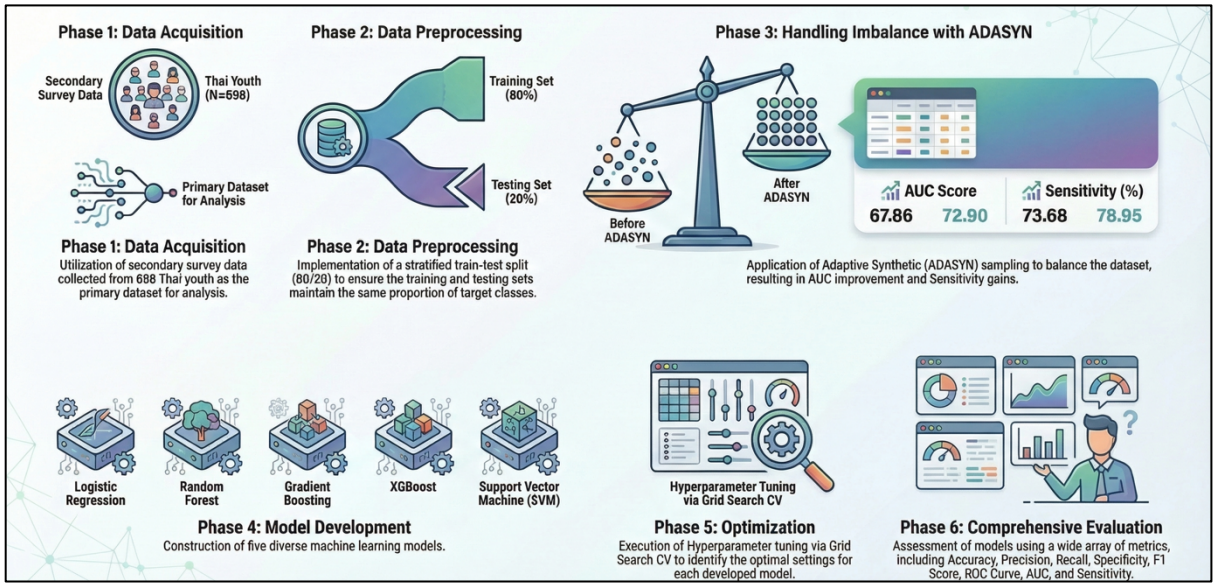

Fig. 1 Predicting Electronic Cigarette Use among Thai Youth Using Machine Learning

Overview of the prediction of electronic cigarette use among Thai youth using machine learning techniques. The framework includes data preprocessing, feature selection using LASSO, handling class imbalance with ADASYN, and model development using five machine learning algorithms. Model performance was evaluated using multiple metrics, with the random forest model selected as the final model based on overall predictive performance.

Result

The sample consisted of 698 individuals aged 15–24 years. Secondary data were obtained from the research titled *"Impact of Mental Health on Alcohol Consumption Behavior Among Adolescents in Bangkok"* conducted by Supaknapha et al. (2024). Five classification approaches were employed: Logistic Regression, Support Vector Machine (SVM), Random Forest, Gradient Boosting Machines (GBM), and Extreme Gradient Boosting (XGBoost). Model performance was evaluated to identify the most effective method for predicting alcohol drinking behavior based on mental health and related factors.

Table 1 Characteristics

	Total	Smoke	No Smoke	*p*
Gender				<0.001*
- Male	274 (39.3)	54 (57.4)	220 (36.4)	
- Female	424 (60.7)	40 (42.6)	384 (63.6)	
Income				0.005*
- Income: 1-10,000 THB	372 (53.3)	36 (38.3)	336 (55.6)	
- Income: 10,001-20,000 THB	243 (34.8)	41 (43.6)	302 (33.4)	
- Income: More Than 20,001 THB	83 (11.9)	17 (18.1)	66 (10.9)	
Agegroup				0.077
- Age 15-19	186 (26.6)	18 (19.1)	168 (27.8)	
- Age More than 20	512 (73.4)	76 (80.9)	436 (72.2)	
Education Level				0.1533
- Below bachelor's degree	175 (25.1)	16 (17.0)	159 (26.3)	
- Bachelor's degree	510 (73.1)	76 (80.9)	434 (71.9)	
- Above bachelor's degree	13 (1.9)	2 (2.1)	11 (1.8)	
Alcohol Drinking Experience				<0.001*
- Ever drink alcohol	568 (81.4)	93 (98.9)	475 (78.6)	
- Never drink alcohol	130 (18.6)	1 (1.1)	129 (21.4)	
Coping Strategies for Stress *(Among participants who reported alcohol consumption)*				<0.001*
- Physical exercise	94 (13.5)	14 (14.9)	80 (13.2)	
- Engaging in hobbies (e.g., playing music, handicrafts)	131 (18.8)	13 (13.8)	118 (9.5)	
- Rest and relaxation (e.g., sleeping, massage, meditation)	186 (26.6)	35 (37.2)	151 (25.0)	

- Spending time with friends or family	107 (15.3)	22 (23.4)	85 (14.1)	
- Seeking advice or emotional support (talking to others)	50 (7.2)	9 (9.6)	41 (6.8)	
- Never drink alcohol	130 (18.6)	1 (1.1)	129 (21.4)	
Living Arrangement				0.119
- Own house	200 (28.7)	20 (21.3)	180 (29.8)	
- Living with family	140 (20.1)	15 (16.0)	125 (20.7)	
- Rented dormitory / apartment / condominium / house	319 (45.7)	52 (55.3)	267 (44.2)	
- University dormitory	39 (5.6)	7 (7.4)	32 (5.3)	
Intention and Status of Alcohol Drinking				<0.001*
- Wants to try drinking alcohol	27 (3.9)	0 (0.0)	27 (4.5)	
- Does not want to drink alcohol	103 (14.8)	1 (1.1)	102 (16.9)	
- Alcohol drinkers	568 (81.4)	93 (98.9)	475 (78.6)	

The Pearson Chi-Square test is considered statistically significant when the P-value $\leq$ 0.05

A preliminary analysis of the general characteristics of the sample (N = 698) revealed that 94 participants (13.5%) were e-cigarette users, while 604 participants (86.5%) had never used e-cigarettes. Among e-cigarette users, 54 were male (57.4%) and 40 were female (42.6%). Regarding monthly income, 36 participants (38.3%) reported an average income of less than 10,000 THB, 41 participants (43.6%) reported an income between 10,001 and 20,000 THB, and 17 participants (18.1%) reported an income exceeding 20,000 THB.

In terms of age distribution, 18 participants (19.1%) were aged between 15 and 19 years, while the majority, 76 participants (80.9%), were aged 20 years or older. With respect to educational attainment, 16 participants (17.0%) had an education level below a bachelor's degree, 76 participants (80.9%) were studying at the bachelor's degree level, and 2 participants (2.1%) had an education level higher than a bachelor's degree.

Regarding living arrangements, 20 participants (21.3%) lived in their own house, 15 participants (16.0%) lived with their family, 52 participants (55.3%) rented accommodation outside the university, and 7 participants (7.4%) resided in university dormitories. Concerning alcohol consumption experience, 93 participants (98.9%) reported having consumed alcohol, whereas only 1 participant (1.1%) reported never having consumed alcohol. In terms of intention to drink alcohol, none of the participants reported wanting to try drinking alcohol (0%), 1 participant (1.1%) reported not wanting to drink alcohol, and 93 participants (98.9%) were classified as alcohol drinkers.

Table 2 Comparison table of model performance using Original data and data processed with Adaptive Synthetic Sampling (ADASYN) and Cross-Validation (CV).

Model	Original (%)						With ADASYN and CV (%)					
	Accuracy	Precision	Sensitivity	Specificity	F1-Score	AUC	Accuracy	Precision	Sensitivity	Specificity	F1-Score	AUC
LR	65.00	25.00	78.95	62.81	37.97	72.59	72.86	26.82	57.89	75.21	36.67	69.99
RF	62.14	22.58	73.68	60.33	34.57	67.86	69.29	27.78	78.95	67.77	41.10	72.90
GBM	87.14	54.55	31.58	95.87	40.00	67.81	70.00	25.53	63.16	71.07	36.36	67.64
XGBoost	79.29	33.33	52.63	83.47	40.81	65.00	75.71	29.73	57.89	78.51	39.29	68.51
SVM	62.14	23.44	78.94	59.50	36.14	71.34	86.42	50.00	31.58	95.04	38.71	70.68

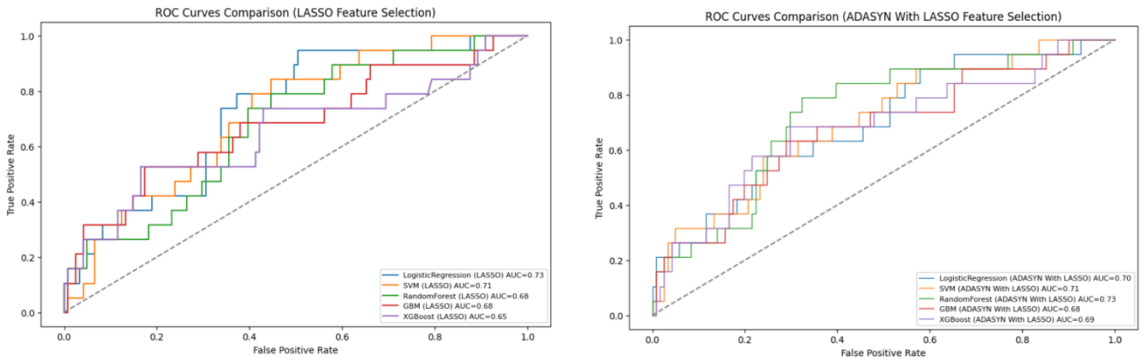

Fig. 2 ROC Curves Comparison (LASSO Feature Selection) with Area Under the Curve (AUC) for the Original data and the ADASYN and cross-validated data.

For the original dataset, the performance metrics of different machine learning models are as follows: Logistic Regression: Accuracy = 65.00, Precision = 25.00, Sensitivity = 78.95, F1-Score = 37.97, Specificity = 62.81, AUC = 0.73, Random Forest: Accuracy = 62.14, Precision = 22.58, Sensitivity = 73.68, F1-Score = 34.57, Specificity = 60.33, AUC = 0.68, Gradient Boosting Machines: Accuracy = 87.14, Precision = 54.55, Sensitivity = 31.58, F1-Score = 40.00, Specificity = 95.87, AUC = 0.68, XGBoost: Accuracy = 79.29, Precision = 33.33, Sensitivity = 52.63, F1-Score = 40.81, Specificity = 83.47, AUC = 0.65, Support Vector Machines: Accuracy = 62.14, Precision = 23.44, Sensitivity = 78.94, F1-Score = 36.14, Specificity = 59.50, AUC = 0.71

After applying Adaptive Synthetic Sampling (ADASYN) and Cross-Validation (CV), the models exhibited the following performance: Logistic Regression: Accuracy = 72.86, Precision = 26.82, Sensitivity = 57.89, F1-Score = 36.67, Specificity = 75.21, AUC = 0.70, Random Forest: Accuracy = 69.29, Precision = 27.78, Sensitivity = 78.95, F1-Score = 41.10, Specificity = 67.77, AUC = 0.73, Gradient Boosting Machines: Accuracy = 70.00, Precision = 25.53, Sensitivity = 63.16, F1-Score = 36.36, Specificity = 71.07, AUC = 0.68, XGBoost: Accuracy = 75.71, Precision = 29.73, Sensitivity = 57.89, F1-Score = 39.29, Specificity = 78.51, AUC = 0.69, Support Vector Machines: Accuracy = 86.42, Precision = 50.00, Sensitivity = 31.58, F1-Score = 38.71, Specificity = 95.04, AUC = 0.71

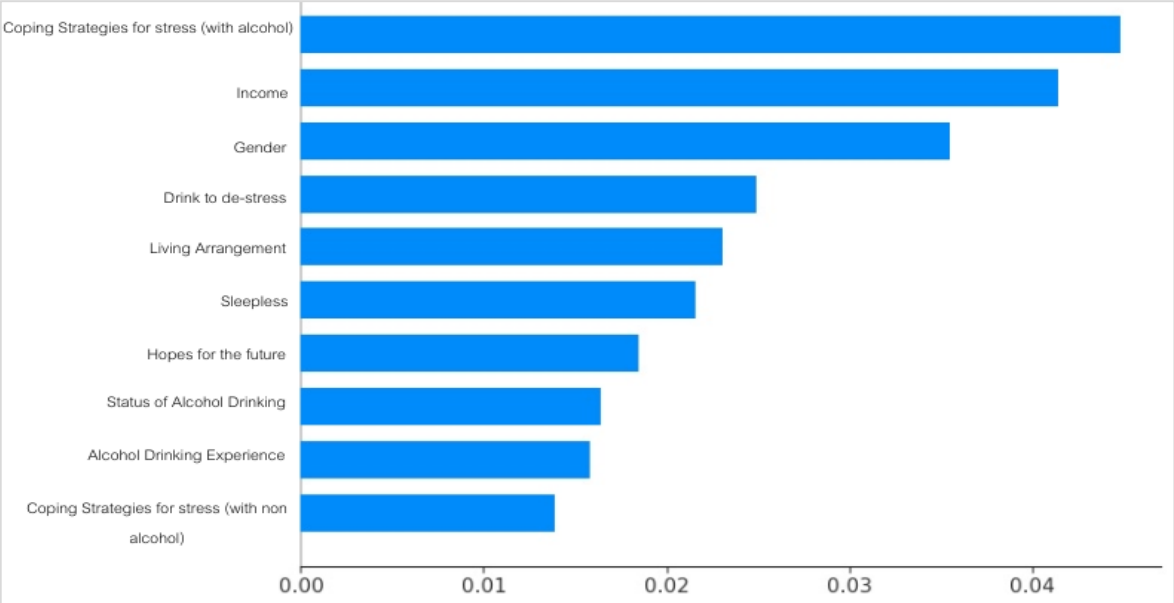

Fig. 3 SHAP impact bar plot of feature contributions for the Random Forest model using ADASYN and cross-validation.

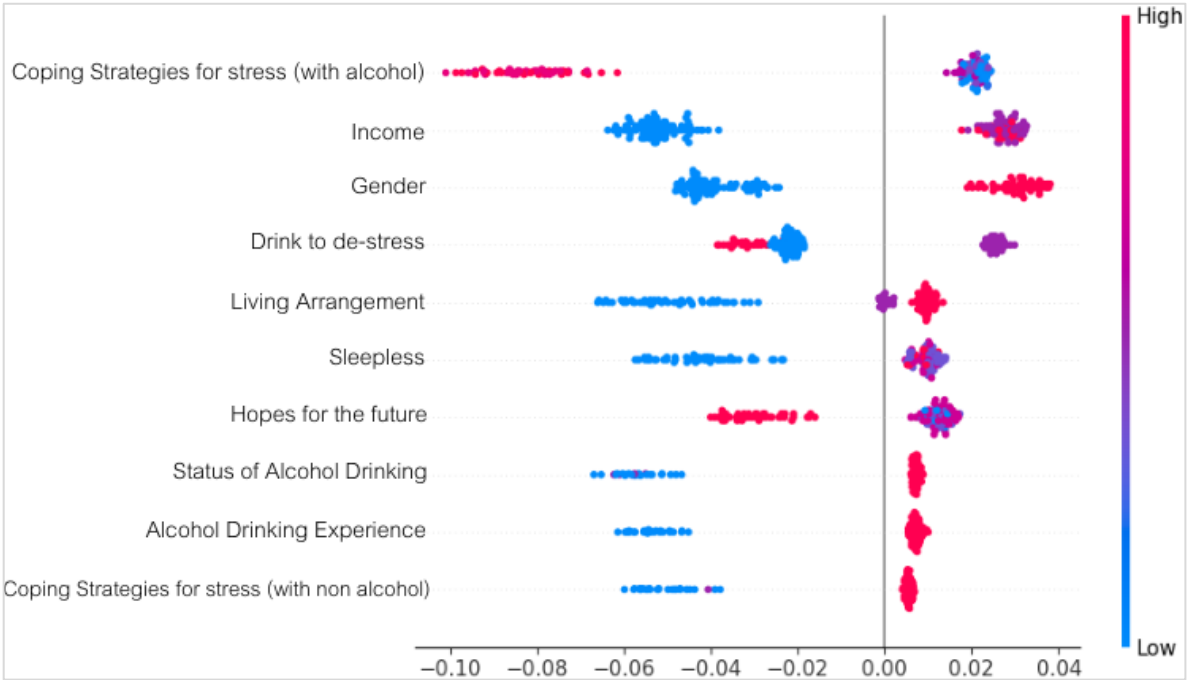

Fig. 4 SHAP summary plot of feature contributions for the Random Forest model using ADASYN and cross-validation.

From SHAP analysis, it was found that the variable "method of problem-solving when stressed" (for those who have consumed alcohol) was the most important variable in predicting e-cigarette use, followed by the respondent's income. Both of these variables had the greatest impact on the model's predictions. The next most important variables were gender, alcohol consumption when stressed (for those who have consumed alcohol), Accommodation, many expectations about the future, sleep problems, desire to try alcoholic beverages, and alcohol consumption experience, which decreased in importance in that order. This indicates that these variables still play a role in explaining the model's predictions but have less influence than the main variables.

Result and Discussion

The analysis of e-cigarette use behavior among Thai youth using machine learning models. The results showed that the Random Forest model provided the best predictive performance when combined with LASSO feature selection, ADASYN data balancing and 5-fold stratified cross-validation, leading to an improvement in the F1-score. The most important factors related to e-cigarette use were coping strategies for dealing with stress, especially among alcohol drinkers, and respondents' income.

The findings are consistent with previous research by Shi et al. (2022), which also reported that Random Forest performed well in predicting e-cigarette use among youth. Although the predictive performance in this study was lower, likely due to a more severe class imbalance and differences in the study population, both studies highlight the importance of behavioral, socioeconomic, and mental health–related factors. Overall, the results suggest that Random Forest is a suitable model for this type of analysis, while future studies may improve prediction accuracy through better data preparation and model development.

Conclusion

The analysis of e-cigarette use behavior among Thai youth using machine learning models found that the Random Forest model achieved the highest performance when combined with LASSO feature selection, ADASYN data balancing, and cross-validation, leading to a significant improvement in F1-score. The most important factors were coping strategies for dealing with stress (among alcohol drinkers) and respondents' income. However, model performance may vary across different datasets and contexts, and further improvements in data preparation and algorithm development may enhance prediction accuracy in future studies.

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